

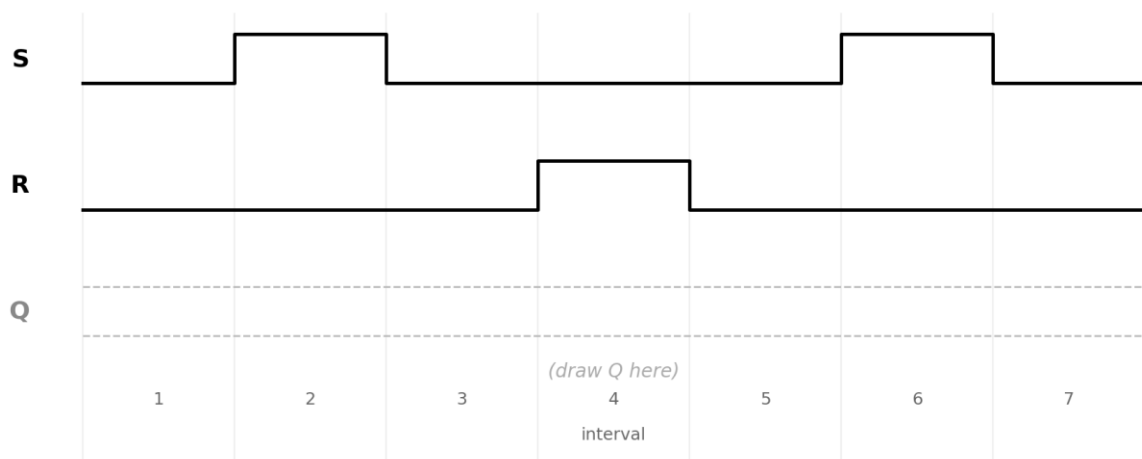
Tutorial 6 — Latches, Flip-Flops, and Sequential Circuit Analysis

Q1 — Complete the Timing Diagrams

Question (take home, no discussion in class needed)

(a) SR Latch

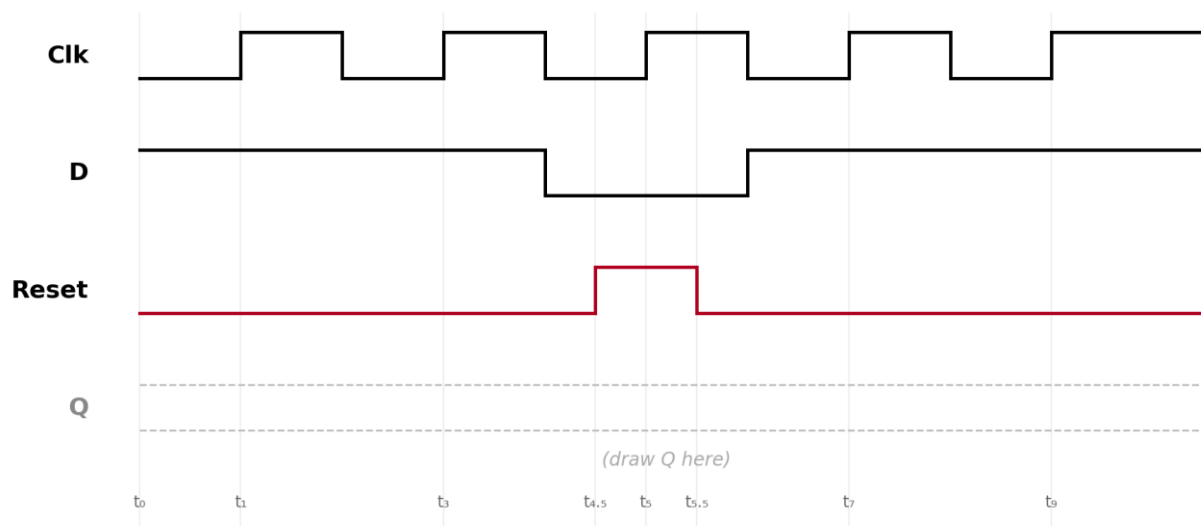
You are given a basic SR latch (NOR-gate based, no control input). Initially, $Q = 0$. The inputs S and R change as shown below.



Complete the timing diagram by drawing Q across all seven intervals, starting from the given initial value. Label each interval with the latch operation taking place (set / reset / hold).

(b) D Flip-Flop with Asynchronous Reset

You are given a positive-edge-triggered D flip-flop with an asynchronous (direct) Reset input, which forces $Q = 0$ immediately whenever asserted, regardless of the clock. Q starts at 0.

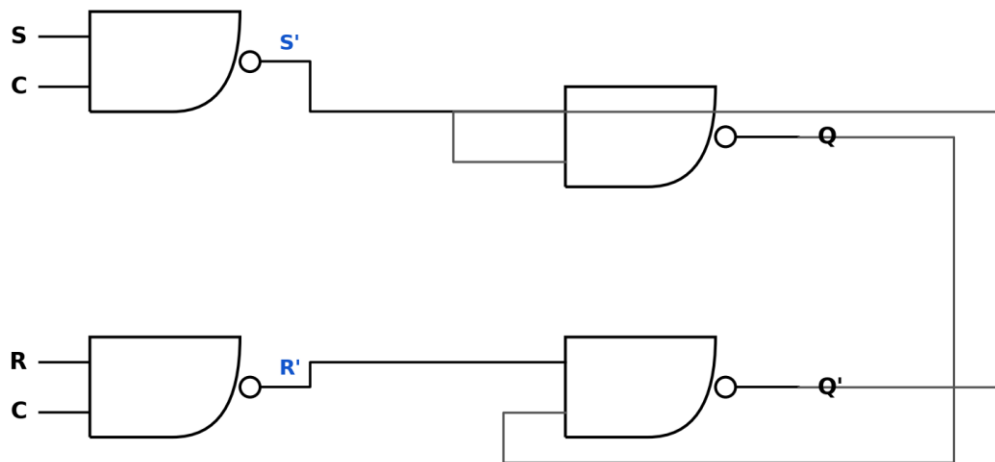


Complete the timing diagram by drawing Q across the full timeline. Pay particular attention to what happens around $t_{4.5}$ – $t_{5.5}$, and to the clock edge at t_5 , which falls inside the reset window.

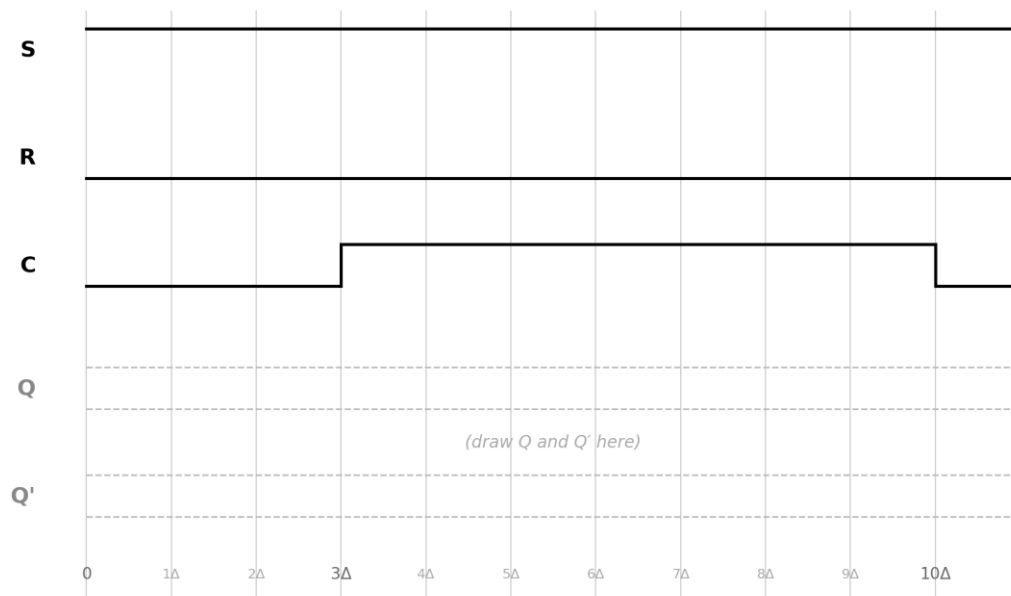
Q2 — Gated SR Latch: Propagation Delay and Settling Time

Question (discuss in class)

Consider the SR latch with a control input C. Assume every gate has the same propagation delay, Δ .



S and R are held fixed at $S = 1$, $R = 0$ throughout. C, along with Q and Q' (which start at $Q = 0$, $Q' = 1$ before $t = 0$), are shown below. Complete the timing diagram for Q and Q' , showing them as unchanged until enough gate delays have elapsed for a change to actually propagate through the circuit. Mark the exact time (in terms of Δ) at which Q first begins to change, and the time at which it has fully stabilized. What happens to Q when C falls back to 0 at $t = 10\Delta$ — does it change at all? Why or why not?



Q3 — Using Latches to Manipulate Signals

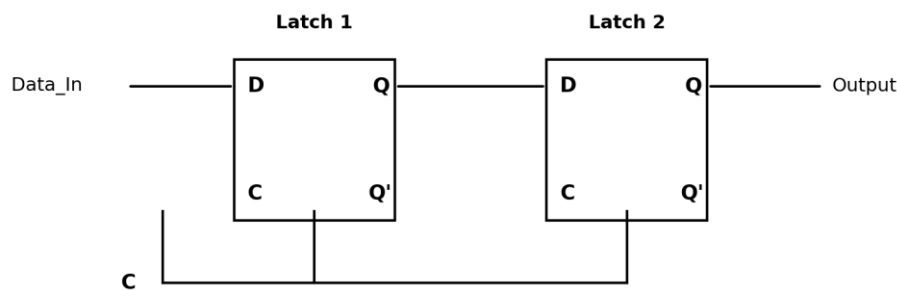
Question (discuss in class)

(a)

Using the same gated SR latch from Q2 (gate delay Δ per gate), suppose C is pulsed high for a duration w to let a new value propagate through, then dropped low again to latch that value in. What is the minimum value of w for the circuit to work correctly? What goes wrong if the pulse is shorter than this — describe the failure concretely, in terms of what Q ends up holding.

(b)

Now consider the circuit below: two identical latches, sharing a single common control line C, with the output of Latch 1 feeding directly into the input of Latch 2. This is meant to work as a 1-bit shift register: on each C pulse, Data_In should move into Latch 1, and whatever Latch 1 was holding before this pulse should move into Latch 2 — a single-stage shift, once per pulse.



Is there a maximum safe duration for this C pulse, beyond which the circuit stops behaving this way? If so, state what goes wrong and why. If not, explain why not.

(c)

How is this entire class of problem — having to worry about a range of valid pulse widths at all — avoided by using edge-triggered flip-flops instead of latches?

Q4 — Design a JK Flip-Flop Using a T Flip-Flop

Question (discuss in class)

You are given a T flip-flop and unlimited basic gates. Design a circuit that behaves as a JK flip-flop — find T as a function of J, K, and the current state $Q(t)$.

Build a table of what T must be for every combination of J, K, and $Q(t)$, then simplify.

Q5 — A New Flip-Flop Type: Compare Two Implementations

Question (optional, discuss if time permits, else take home)

Consider a flip-flop with two inputs, A and B, and the following behavior:

A	B	$Q(t+1)$
0	0	0 (reset)
0	1	$Q'(t)$ (toggle)
1	0	$Q'(t)$ (toggle)
1	1	1 (set)

Design this flip-flop two different ways:

- (a) Using a D flip-flop as the building block — find D as a function of A, B, and $Q(t)$.

- (b) Using a JK flip-flop as the building block — find J and K as functions of A, B, and Q(t), using the JK excitation table.

Count the total literals in your simplified expression(s) for each version. Which implementation is cheaper, and why? What is it about the JK excitation table specifically that produces this difference?

Q6 — Analyze This Sequential Circuit

Question (discuss in class)

A sequential circuit uses two T flip-flops, with present state Q_1Q_0 , one input X, and one output Z. The flip-flop input equations and the output equation are:

$$T_1 = X \cdot Q_0$$

$$T_0 = X$$

$$Z = X \cdot Q_1 \cdot Q_0$$

(a)

Construct the complete state table: present state, input, flip-flop inputs, next state, and output, for every combination of Q_1Q_0 and X.

(b)

Draw the state diagram.

(c)

In one or two sentences, describe in plain language what this circuit does.